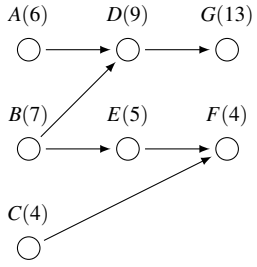
Name: _____ score:_____ / 10

There are 10 points possible on this quiz. No aids (book, notes, etc.) are permitted. You may use a non-programmable calculator. **Show all work and supporting calculations for full credit. Explain how you get your answers.**

1. Consider the following directed graph.



(a) (1 pt) What tasks need to be completed before task D can start?

(b) (1 pt) What tasks are available to be assigned at the beginning of the project?

(c) (1 pt) What is the critical path?

(d) (1 pt) What is the critical time? _____

2. Suppose you have a project consisting of the seven tasks described in the following table.

   (a) (2 pts) Create a project digraph with vertices labeled. Draw it next to the table.

label/task	time	dependence
A	7	
B	3	
C	4	
D	3	A
E	8	A, B
F	2	E

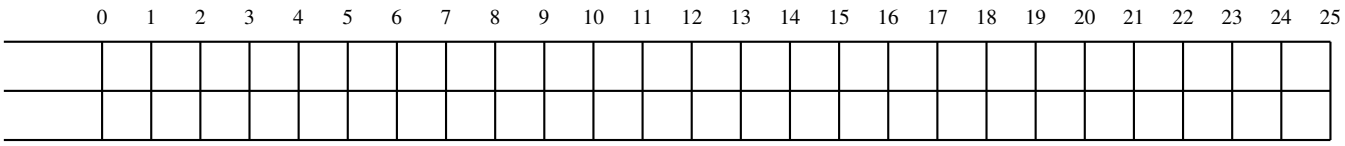
   (b) (3 pts) Schedule the project with 2 processors using priority list

   $$A \quad B \quad C \quad D \quad E \quad F$$

   Make sure you write clearly so that someone else can read your schedule!

   The finishing time is _____ and the total idle time is _____.

time	
ready	
done	



   (c) Create a priority list for this project based on the Decreasing-Time algorithm.